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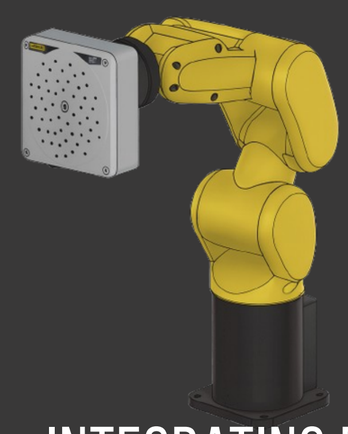
ATU Galway · 2025/2026 · Final Year



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Atlantic Technological University

ROBOTIC LEAK DETECTION SYSTEM

INTEGRATING ROBOTICS · PLC AUTOMATION · SCADA VISUALISATION

PROJECT OVERVIEW

Developed in partnership with **Trane Technologies**, this system automates acoustic leak detection of brazed copper HVAC joints. Undetected leaks cause product failures, costly rework, and wasted energy across global manufacturing facilities.

A **FANUC LR MATE 200iD** robot positions a **Fluke SV600** acoustic imager across 3 scan positions, orchestrated by an **Allen-Bradley ControlLogix PLC** and **Ignition SCADA**. All results are logged to MySQL with full traceability via QR-decoded serial numbers.

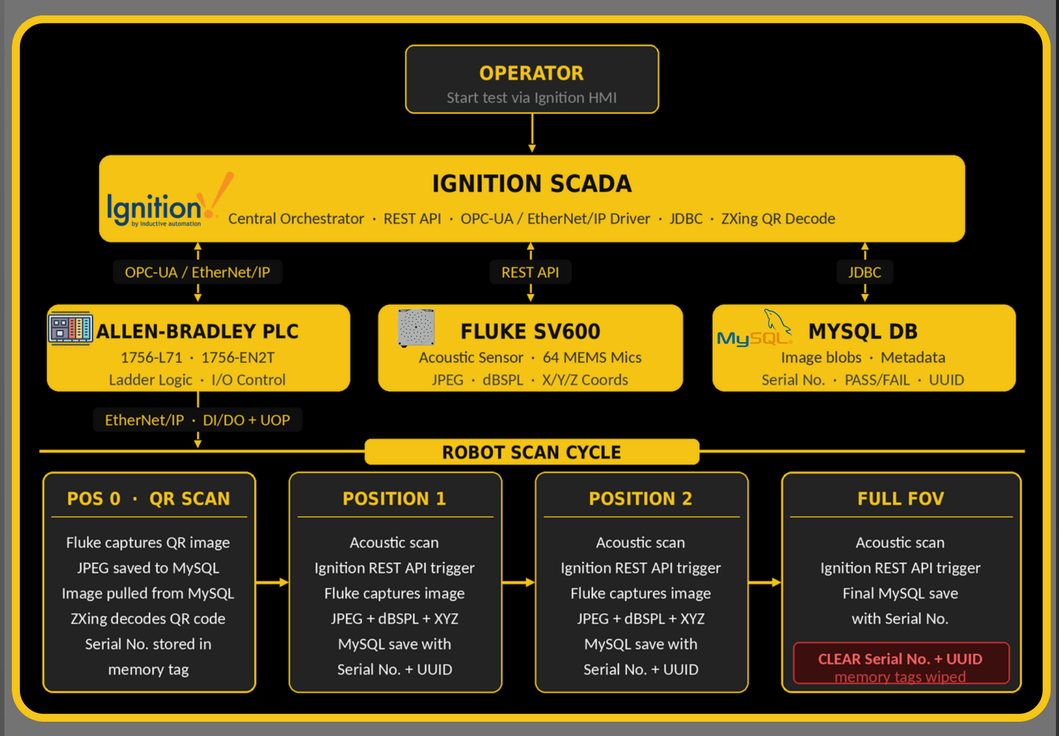
TECHNOLOGIES

- FANUC LR MATE 200iD
- ALLEN-BRADLEY 1756-L71
- FLUKE SV600
- IGNITION SCADA
- ETHERNET/IP
- MYSQL DATABASE
- REST API
- OPC-UA
- ZXING (JAVA)
- STUDIO 5000
- JYTHON

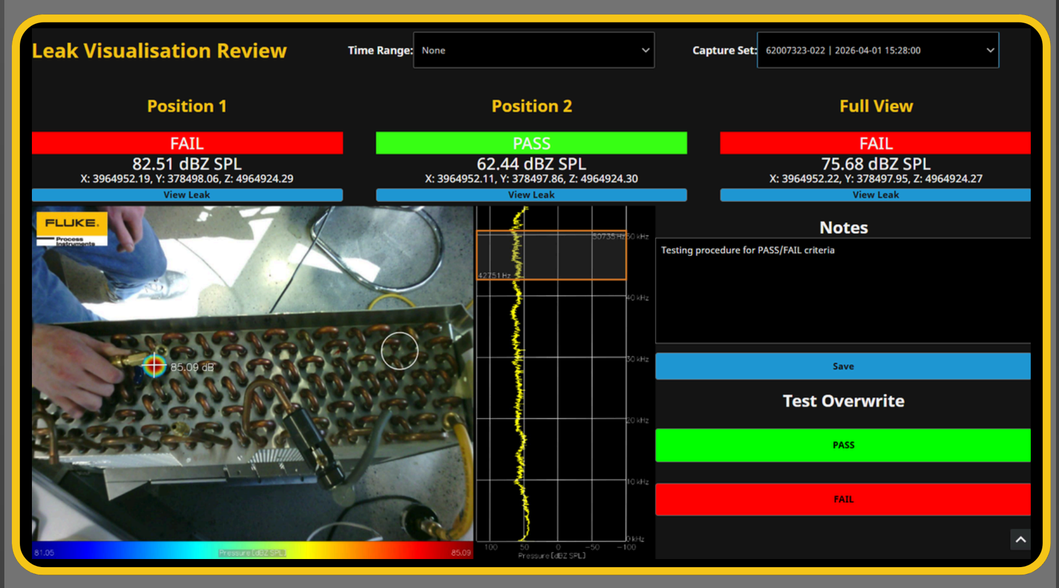
SYSTEM WORKFLOW

1. Operator presses Start on Ignition HMI
2. PLC checks robot is safe - enables UOP signals via EtherNet/IP
3. Robot moves to QR scan position - Fluke captures and ZXing decodes serial number into memory tag
4. Robot moves to Pos 1, 2 and Full FOV - Ignition REST API captures SV600 heatmap image at each position
5. JPEG, dBSPL, XYZ and PASS/FAIL saved to MySQL - operator reviews and can override results on Ignition dashboard.

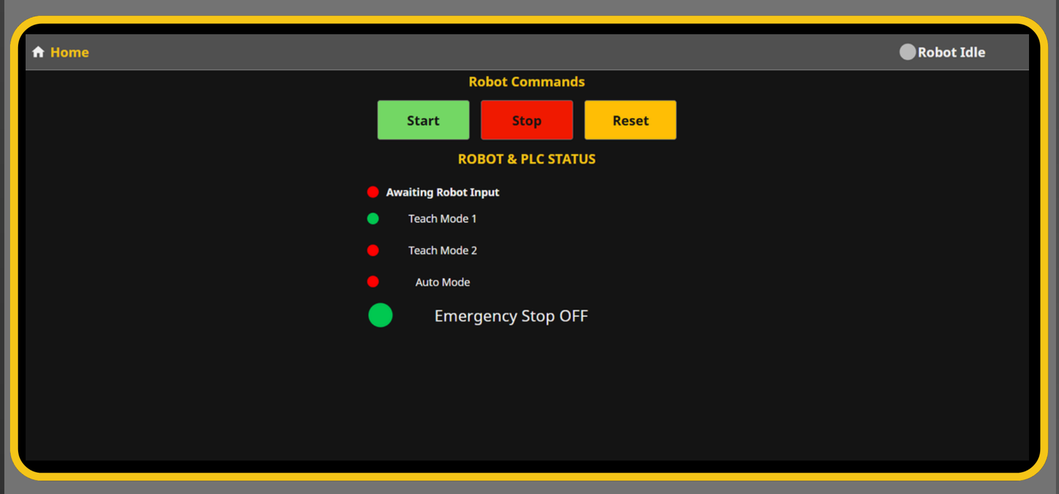
SYSTEM ARCHITECTURE



RESULTS SCREEN



COMMANDS SCREEN



ROBOT SETUP

KEY RESULTS

350MM
Optimal sensor-to-product distance

48-51KHZ
Optimal detection frequency range

AUTOMATED
Full pipeline - Zero manual steps

Automated PASS/FAIL classification per position with dB SPL threshold:

PASS FAIL per scan position

QR code decoded on product → serial number logged to MySQL for full traceability across all 3 positions per test cycle.

BEFORE VS AFTER

	BEFORE	AFTER
Inspection	Manual	Automated
Distance	~2000mm	350mm
Frequency	Broadband	48-51kHz
Traceability	None	QR → MySQL
Steps	Manual	Zero

CONCLUSION

The system successfully delivers a fully automated end-to-end detection pipeline - from robot motion through acoustic imaging to PASS/FAIL classification logged in MySQL

Validated on HVAC coil products at Trane Technologies, Galway.

FUTURE WORK

- **LLM Integration**
  Enable operators to query test results and system status in natural language from the Ignition dashboard
- **AI Leak Localisation**
  Train a model to identify the exact coil section where a leak originates from the SV600 acoustic heatmap